

Math 465: Introduction to Combinatorics

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Homework #4 is posted.

You get ONE email to me per semester saying “I was submitting at 11:59 and my Wi-Fi/Gradescope failed” shortly after the HW deadline. Please submit early enough to allow for technical issues!

These slides will be posted on Canvas.

Non-homogeneous linear recurrences

Problem

Let $a_1, \dots, a_k$ be as before. Given a sequence $(b_n) = (b_0, b_1, b_2, \dots)$, find the solution of a **linear non-homogeneous recurrence**

$$(*)' \quad h_n + a_1 h_{n-1} + a_2 h_{n-2} + \dots + a_k h_{n-k} = b_n \quad (n \geq k)$$

satisfying prescribed initial conditions.

Observation

The difference of any two solutions of the recurrence $(*)'$ satisfies the corresponding homogeneous recurrence $(*)$.

Hence it suffices to find one **particular solution** of $(*)'$. The general solution of $(*)'$ is then obtained by adding to this particular solution the general solution of $(*)$.

A particular solution of $(*)'$ can often be found **within the same class of sequences** that (b_n) belongs to. E.g., if b_n is a polynomial in n , try to find h_n that is also a polynomial. If $b_n = b_0 r^n$, try $h_n = h_0 r^n$, etc.

Non-homogeneous linear recurrences: an example

Problem

The sequence $(h_n) = (1, 1, 4, 5, 14, 23, 52, 97, \dots)$ is defined by

$$h_n = h_{n-1} + 2h_{n-2} + (-1)^n,$$

with the initial conditions $h_0 = h_1 = 1$. Find a formula for h_n .

Solution

The characteristic equation is $q^2 - q - 2 = 0$. Its roots are 2 and -1 . In this example, $b_n = (-1)^n$. Since -1 is a root of the characteristic equation, there is no solution of the form $h_n = c \cdot (-1)^n$. Let us try a slight modification of the latter, namely $h_n = c n (-1)^n$:

$$\begin{aligned} h_n - h_{n-1} - 2h_{n-2} &= c [n(-1)^n - (n-1)(-1)^{n-1} - 2(n-2)(-1)^{n-2}] \\ &= c (-1)^n [n + n - 1 - 2n + 4] \\ &= 3c (-1)^n, \end{aligned}$$

so $c = \frac{1}{3}$ works! Thus $h_n = c_1 2^n + c_2 (-1)^n + \frac{1}{3} n (-1)^n$. Fitting c_1 and c_2 to the initial conditions, we obtain $h_n = \frac{7}{9} 2^n + \frac{2}{9} (-1)^n + \frac{1}{3} n (-1)^n$.

Characterization of sequences given by polynomials (1)

We next discuss an important application of the theory of (homogeneous) linear recurrences.

Corollary [see next slide for the proof]

The general solution (h_n) of the linear recurrence

$$\sum_{i=0}^k (-1)^i \binom{k}{i} h_{n-i} = 0$$

is a polynomial in n of degree $\leq k - 1$.

Example ($k = 4$)

The solutions of the linear recurrence

$$h_n - 4h_{n-1} + 6h_{n-2} - 4h_{n-3} + h_{n-4} = 0$$

are given by $h_n = an^3 + bn^2 + cn + d$.

A simpler example: $k = 2$, arithmetic progressions.

Characterization of sequences given by polynomials (2)

Corollary

The general solution (h_n) of the linear recurrence

$$\sum_{i=0}^k (-1)^i \binom{k}{i} h_{n-i} = 0$$

is a polynomial in n of degree $\leq k - 1$.

Proof

In this case, the characteristic equation is

$$\sum_{i=0}^k (-1)^i \binom{k}{i} q^{k-i} = (q - 1)^k = 0.$$

Hence the general solution is $h_n = C(n) \cdot 1^n$ where $C(n)$ is a polynomial of degree $< k$.

Difference tables

For an infinite sequence (h_n) , the associated **difference sequence** is the sequence $(h_{n+1} - h_n)$:

$$\begin{array}{ccccccc} h_0 & & h_1 & & h_2 & & h_3 & & h_4 & & \dots \\ & h_1 - h_0 & & h_2 - h_1 & & h_3 - h_2 & & h_4 - h_3 & & \dots \end{array}$$

Iterating, we obtain a **difference table**, shown below for $h_n = \binom{n}{3}$:

$$\begin{array}{cccccccccc} 0 & 0 & 0 & 1 & 4 & 10 & 20 & 35 & \dots \\ & 0 & 0 & 1 & 3 & 6 & 10 & 15 & \dots \\ & & 0 & 1 & 2 & 3 & 4 & 5 & \dots \\ & & & 1 & 1 & 1 & 1 & 1 & \dots \\ & & & & 0 & 0 & 0 & 0 & \dots \\ & & & & & 0 & 0 & 0 & \dots \end{array}$$

Entries of a difference table

Proposition

The k 'th differences of a sequence (h_n) are given by

$$\sum_{i=0}^k (-1)^i \binom{k}{i} h_{n-i},$$

for $n \geq k$.

Proof

h_0	h_1	h_2	h_3	h_4
$h_1 - h_0$	$h_2 - h_1$	$h_3 - h_2$	$h_4 - h_3$	
$h_2 - 2h_1 + h_0$	$h_3 - 2h_2 + h_1$	$h_4 - 2h_3 + h_2$		
$h_3 - 3h_2 + 3h_1 - h_0$	$h_4 - 3h_3 + 3h_2 - h_1$			

Characterization of sequences given by polynomials (3)

Theorem

A sequence $h = (h_n)$ is given by a polynomial in n of degree $\leq k - 1$ if and only if all of its k 'th differences vanish.

Example: $h_n = n^3$

0	1	8	27	64	125	216	343	...
	1	7	19	37	61	91	127	...
		6	12	18	24	30	36	...
			6	6	6	6	...	
			0	0	0	0	...	
				0	0	0	...	

Characterization of sequences given by polynomials (4)

Theorem

A sequence $h = (h_n)$ is given by a polynomial in n of degree $\leq k - 1$ if and only if all of its k 'th differences vanish.

Proof #1

The vanishing of the k 'th differences is equivalent to the equation

$$\sum_{i=0}^k (-1)^i \binom{k}{i} h_{n-i} = 0.$$

Solutions of this equations are given by polynomials of degree $< k$.

Proof #2

It suffices to show [why?] that h_n is a polynomial in n of degree d if and only if $h_{n+1} - h_n$ is a polynomial in n of degree $d - 1$.

One direction follows by observing that $(n+1)^a - n^a$ is a polynomial of degree $a - 1$. For the opposite direction, use summation formulas.

Extrapolation of polynomials

Recall: a sequence (h_n) is given by a polynomial in n of degree $< k$ if and only if its k 'th differences vanish. Application: [extrapolation](#).

Problem

Extrapolate a cubic polynomial $f(x)$ given the following values:

x	a	$a + \delta$	$a + 2\delta$	$a + 3\delta$
$f(x)$	1	-2	-5	4

Solution

Let us build a difference table for the sequence $h_n = f(a + n\delta)$.

The bottom entry 12 is [replicated in the bottom row \[why?\]](#).

Then the table is [constructed bottom-up, column by column](#).

1	-2	-5	4	37	106
-3	-3	9	33	69	
0	12	24	36		
12	→ 12	→ 12			

Determining a polynomial from its values at integer points

1		-2		-5		4		37		106
	-3		-3		9		33		69	
		0		12		24		36		
			12		12		12			

We did not need a formula for the cubic polynomial $f(a+n\delta)$ in order to extrapolate it. Such a formula can be written using the leftmost entries in the rows of the difference table:

$$\begin{aligned}h_n &= 1 \cdot \binom{n}{0} - 3 \cdot \binom{n}{1} + 0 \cdot \binom{n}{2} + 12 \cdot \binom{n}{3} \\&= 1 - 3n + 2n(n-1)(n-2) \\&= 2n^3 - 6n^2 + n + 1.\end{aligned}$$

Discrete Taylor's formula

Let $\hat{h}_0, \hat{h}_1, \hat{h}_2, \dots$ be the leftmost entries in the difference table for (h_n) . They are the analogues of the derivatives of a function at 0.

Proposition/Exercise

The leftmost entries of the difference table are the coefficients in the expansion of h_n in terms of binomial coefficients $\binom{n}{m}$:

$$h_n = \sum_m \hat{h}_m \binom{n}{m} = \sum_m \frac{\hat{h}_m}{m!} \cdot (n)_m$$

Discrete Taylor's formula: an example

Example: $h_n = n^3$

0	1	8	27	64	125	216	343	...
	1	7	19	37	61	91	127	...
		6	12	18	24	30	36	...
			6	6	6	6	...	
			0	0	0	0	...	
				0	0	0	...	

$$n^3 = \binom{n}{1} + 6\binom{n}{2} + 6\binom{n}{3}.$$

Catalan numbers

Definition

The *Catalan numbers* C_n are defined by

$$C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{n!(n+1)!} = \frac{1}{2n+1} \binom{2n+1}{n}.$$

n	0	1	2	3	4	5	6	7	8	9
C_n	1	1	2	5	14	42	132	429	1430	4862

These numbers enumerate many families of combinatorial objects.

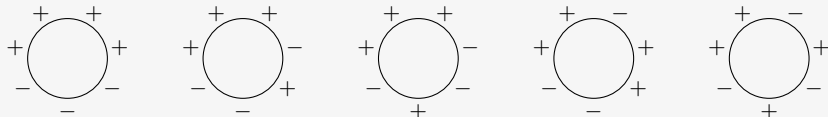
Catalan numbers and circular arrangements

n	0	1	2	3	4	5	6	7	8	9
C_n	1	1	2	5	14	42	132	429	1430	4862

Theorem

The number of circular arrangements of $n + 1$ pluses and n minuses, considered up to rotation, is the Catalan number C_n .

Example: $n = 3$



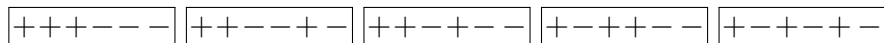
Proof

To obtain the formula $\frac{1}{2n+1} \binom{2n+1}{n}$ for the number in question, use the Division Principle together with the fact that $\gcd(n, n+1) = 1$.

Ballot sequences

Definition

A **ballot sequence** of length $2n$ is a sequence of n pluses and n minuses in which each initial segment contains at least as many pluses as minuses.



Interpreting each $+$ as 1 and each $-$ as -1 , a ballot sequence is a sequence $(a_1, \dots, a_{2n})$ such that

- each component a_i is equal to either 1 or -1 ,
- $a_1 + \dots + a_{2n} = 0$, and
- each partial sum $a_1 + \dots + a_i$ is nonnegative.

Theorem

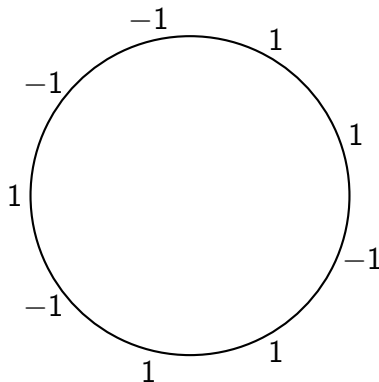
The number of ballot sequences of length $2n$ is C_n .

The proof will rely on the following lemma.

Circular arrangements of ± 1 's with total sum 1

Lemma

In any circular arrangement of ± 1 's whose sum is equal to 1, there is exactly one location starting from which all clockwise partial sums are positive.



Circular arrangements of ± 1 's with total sum 1

Lemma

In any circular arrangement of ± 1 's whose sum is equal to 1, there is exactly one location starting from which all clockwise partial sums are positive.

Proof of uniqueness

If there are two such locations, then the circle breaks into two segments, each summing to ≥ 1 ; so the total is ≥ 2 , a contradiction.

Proof of existence

Start someplace, find a non-positive partial sum. (If it doesn't exist, we are done.) Start at its end, find a non-positive partial sum, etc. Eventually we must step into a place that has already been visited (after some number of revolutions). It follows that the sum of all entries is ≤ 0 , a contradiction.

Counting ballot sequences

Lemma

In any circular arrangement of ± 1 's whose sum is equal to 1, there is exactly one location starting from which all clockwise partial sums are positive. Equivalently, there is exactly one entry whose removal leaves a ballot sequence.

This yields a bijection between ballot sequences of length $2n$ and circular arrangements of $n + 1$ pluses and n minuses.

Since the number of such arrangements is C_n , we conclude:

Theorem

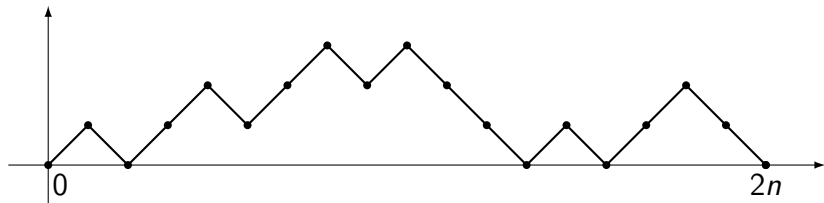
The number of ballot sequences of length $2n$ is C_n .

Dyck paths

Definition

A **Dyck path** of length $2n$ is a path in the coordinate plane which

- connects the points $(0, 0)$ and $(2n, 0)$,
- consists of steps with displacement $(1, 1)$ or $(1, -1)$, and
- is entirely contained in the upper half-plane $y \geq 0$.



Dyck paths can be interpreted as trajectories of a random walk on $\mathbb{Z}$ which begin and end at the origin and stay within the nonnegative half-axis.